

Accident Detection and Alert System for Two-Wheelers using ESP32

MINI PROJECT REPORT

Submitted by

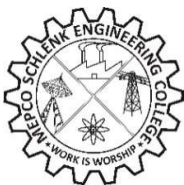
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MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI
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BONAFIDE CERTIFICATE

Certified that this project report titled “**ACCIDENT DETECTION AND ALERT SYSTEM FOR TWO-WHEELERS USING ESP32**” is the bonafide work of **Mr. V.B. BALAPRASHANNA (REG.No:9517202302016), Mr. G. DHARUN AKASH (REG.NO:9517202302027) and Mr. A. AUGUSTIN ILANGO (REG.NO:9517202302012)** who carried out the mini-project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents an accident detection and alert system for two-wheelers using the ESP32 microcontroller. Road accidents often result in delayed medical assistance, increasing the risk of serious injuries and fatalities. The system employs sensors such as an accelerometer and gyroscope to continuously monitor the movement and orientation of the vehicle. Sudden impacts or abnormal tilts are identified as potential accidents. Once detected, the ESP32 processes the data in real time and triggers an alert mechanism. A GPS module is used to determine the exact location of the accident, which is then sent to predefined emergency contacts through GSM. A manual override feature is included to minimize false alerts. The system is designed to be compact, cost-effective, and suitable for real-world implementation. By ensuring quick detection and immediate alert transmission, the proposed system helps reduce response time and improves the chances of timely medical assistance and survival.

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CHAPTER 1

INTRODUCTION

Road accidents involving two-wheelers have become a major public safety issue, particularly in developing countries where motorcycles are a primary mode of transportation. Due to minimal physical protection, riders are highly exposed to injuries during accidents. According to various studies, a significant number of fatalities occur not only because of the severity of the accident but also due to delays in detecting the incident and providing timely medical assistance. In many situations, especially in remote or less crowded areas, victims are unable to call for help, which further increases the risk of fatal outcomes.

To address this critical problem, there is a growing need for an automated system that can detect accidents in real time and immediately notify emergency contacts. With the rapid advancement of embedded systems, wireless communication, and IoT technologies, it has become feasible to develop smart safety solutions that enhance road safety. This project proposes an Accident Detection and Alert System for two-wheelers using the ESP32 microcontroller, which offers high processing capability along with built-in communication features.

The system utilizes the MPU6050 sensor module, which combines a 3-axis accelerometer and a 3-axis gyroscope, to continuously monitor the motion, vibration, and orientation of the vehicle. By analysing sudden changes in acceleration and tilt, the system can accurately detect unusual conditions that indicate a crash or fall. Once such an event is identified, the ESP32 processes the sensor data and triggers an alert mechanism.

To ensure effective communication, the system incorporates the NEO-6M GPS module to determine the exact location (latitude and longitude) of the accident. This location data is then transmitted through the SIM800L GSM module in the form of an SMS to predefined emergency contacts such as family

members, friends, or medical services. This immediate notification helps in reducing response time and enables quicker rescue operations.

Additionally, the system includes a manual override feature that allows the user to cancel the alert in case of false detection, thereby improving reliability. The overall design of the system is compact, energy-efficient, and cost-effective, making it practical for real-world deployment in two-wheelers without requiring significant modifications.

In conclusion, this project aims to leverage modern technology to improve road safety by providing an efficient and reliable accident detection and alert mechanism. By ensuring rapid communication and timely assistance, the system has the potential to significantly reduce fatalities and enhance the chances of survival for accident victims.

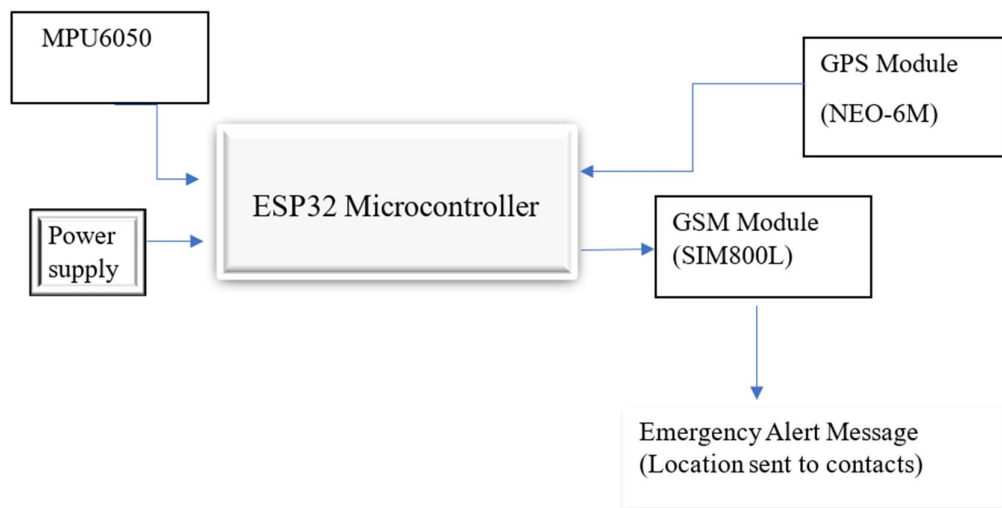


Fig 1.1 Block diagram

CHAPTER 2

LITERATURE REVIEW

Several researchers have proposed accident detection and alert systems to reduce fatalities caused by delayed emergency response. An IoT-based accident detection system using the ESP32 microcontroller was developed to detect crashes in real time using accelerometer data and send alerts along with location details via GPS and communication modules. The system demonstrated that immediate notification to emergency contacts significantly improves rescue response time .

A study on vehicle accident detection using an MPU6050 sensor highlighted the effectiveness of combining accelerometer and gyroscope data to identify sudden changes in motion and tilt. The system used GPS and GSM modules to transmit accident location through SMS, proving that sensor-based detection is reliable and cost-effective for real-time applications .

IoT-based accident alert systems have also been developed using multiple sensors and ESP32 to detect collision, rollover, and fire hazards. These systems integrated GPS modules like NEO-6M and cloud platforms to provide real-time tracking and improve monitoring efficiency .

Earlier systems focused on combining sensors such as accelerometers, pressure sensors, and speed monitoring to accurately determine accident conditions and reduce false alarms. These systems also emphasized sending location-based alerts to emergency services and relatives for faster assistance .

Recent advancements include the use of machine learning algorithms to improve accident detection accuracy in two-wheelers. These methods analyze motion patterns and distinguish between normal riding conditions and actual crashes, thereby minimizing false positives .

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

This system is designed as an embedded system integrating sensing, processing, positioning, and communication modules. The central component of the system is the ESP32 microcontroller, which coordinates all operations and processes data from connected modules.

The system consists of four main hardware components: ESP32, MPU6050 sensor, NEO-6M GPS module, and SIM800L GSM module. The MPU6050, which includes a 3-axis accelerometer and gyroscope, is used to continuously monitor the motion, vibration, and tilt of the vehicle. It is connected to the ESP32 using the I2C communication protocol, enabling real-time data acquisition.

The NEO-6M GPS module is used to determine the precise geographical location of the vehicle. It communicates with the ESP32 via UART and provides latitude and longitude coordinates when required. The SIM800L GSM module is also interfaced using UART and is responsible for sending emergency alert messages to predefined contacts.

The system operates by continuously analyzing data from the MPU6050 sensor. Predefined threshold values for acceleration and tilt are set within the ESP32. When these values exceed the limits, the system identifies it as a potential accident. A verification mechanism, such as a short delay or condition check, is included to reduce false detections caused by normal riding conditions like sudden braking or bumps.

Upon confirming an accident, the ESP32 retrieves the current location from the GPS module and sends an alert message via the GSM module, including the location details. A manual override switch can also be included in the design to cancel false alerts.

Hardware Components

1.ESP32

Specifications

- Operating Voltage: 3.3V
- Dual-core processor with up to 240 MHz clock speed
- Built-in Wi-Fi and Bluetooth
- Multiple GPIO pins (Digital, PWM, ADC, DAC)
- Communication protocols: UART, I2C, SPI
- Low power consumption

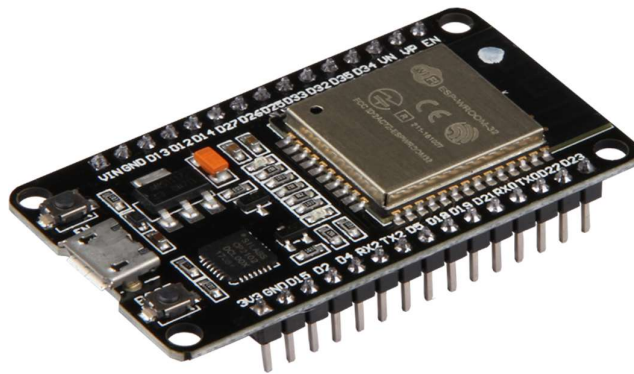


Fig 3.1 ESP32

Connectivity

The ESP32 serves as the central controller and is connected to all other modules in the system. It interfaces with the MPU6050 sensor using the I2C communication protocol through its SDA and SCL pins. The NEO-6M GPS module and SIM800L GSM module are connected to the ESP32 via UART communication, where the TX and RX pins of each module are linked to the corresponding RX and TX pins of the ESP32. The ESP32 is powered using a stable 3.3V supply, and all modules share a common ground connection to ensure proper communication and system stability.

Functional Description

The ESP32 acts as the central processing unit of the system, controlling and coordinating all operations. It continuously receives motion and orientation data from the MPU6050 sensor and analyses it to detect abnormal conditions such as sudden impacts or excessive tilt. When an accident is identified, the ESP32 processes the event and initiates the alert mechanism. It communicates with the GPS module to obtain the current location and then uses the GSM module to send an emergency message to predefined contacts. Thus, the ESP32 integrates sensing, decision-making, and communication functionalities into a single system.

2.MPU6050

Specifications

- Operating Voltage: 3V – 5V
- 3-axis accelerometer and 3-axis gyroscope
- Digital output via I2C protocol
- Built-in Digital Motion Processor (DMP)
- High sensitivity and accuracy

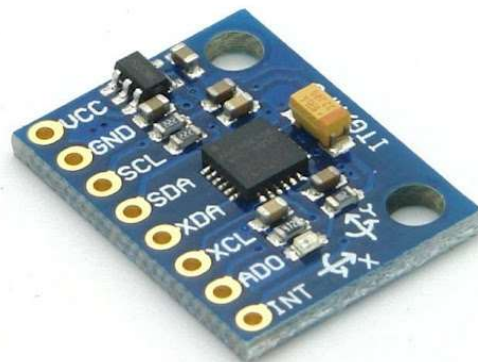


Fig 3.2 MPU6050

Connectivity

The MPU6050 sensor is connected to the ESP32 using the I2C protocol. Its SDA (Serial Data) pin is connected to the SDA pin of the ESP32, and its SCL (Serial Clock) pin is connected to the SCL pin of the ESP32. The sensor is powered through the VCC pin, which can be connected to either 3.3V or 5V depending on the module version, and the GND pin is connected to the common ground. This setup allows continuous transmission of motion and orientation data to the ESP32.

Functional Description

The MPU6050 functions as the motion sensing unit of the system. It combines a 3-axis accelerometer and a 3-axis gyroscope to measure linear acceleration and angular velocity of the two-wheeler. The sensor continuously monitors changes in movement, vibration, and tilt. When a sudden impact, fall, or unusual orientation is detected, it generates corresponding data that is sent to the ESP32. This data plays a crucial role in identifying accident conditions accurately.

3.Battery Holder

Specifications

- Component Type: Battery holder or case
- Supported Battery: Single-cell 3.7V Li-ion (e.g., 18650)
- Material: Durable ABS plastic body
- Terminals: Nickel-plated steel spring contacts
- Output Voltage: 3.7V (for single-cell)
- Connection Leads: Red (positive) and black (negative) wires
- Mounting Type: Screw holes or adhesive base



Fig 3.3 Battery Holder

Functional Description

The battery holder provides a secure mount and power connection for the battery used in the IoT-based monitoring system. It supplies the required voltage to components such as the SIM800L, ensuring stable and continuous power. Its simple design prevents loose connections and enables safe handling of the battery within the system.

4.Bread Board

A bread board is a solderless prototyping platform used to build and test electronic circuits easily without permanent connections. It is typically made of durable ABS plastic with metal contact strips arranged in rows and columns to create electrical connections when components or wires are inserted. The central area is used for interconnecting components like resistors, sensors, and microcontrollers, while the side rails supply power (VCC and GND). In this project, the breadboard is used to connect the ESP32, MPU6050, SIM800L, and NEO-6M, allowing easy circuit assembly, testing, and modification during the development of the accident detection and alert system.

5.Li ion battery



Fig 3.4 Li ion Battery

A Li-ion (Lithium-Ion) battery is a rechargeable power source commonly used in IoT projects for its high energy density and long cycle life. It typically provides a nominal voltage of 3.7V per cell and can be charged up to 4.2V using a charging circuit like the TP4056 module.

Li-ion batteries are lightweight, have low self-discharge rates, and can deliver high current output, making them suitable for powering microcontrollers such as the ESP8266. In this project, the Li-ion battery acts as the main power source, supplying stable DC voltage to all components and ensuring continuous operation even in portable or off-grid conditions.

6.NEO-6M GPS Module

Specifications:

- Operating Voltage: 3.3V – 5V
- Provides latitude, longitude, altitude, and time
- High sensitivity GPS receiver
- UART communication interface
- Accuracy: ~2.5 meters



Fig 3.5 NEO-6M GPS Module

Connectivity

The NEO-6M GPS module is connected to the ESP32 using UART communication. The TX pin of the GPS module is connected to the RX pin of the ESP32, and the RX pin of the GPS module is connected to the TX pin of the ESP32. The module is powered using a 3.3V or 5V supply, and the ground is shared with the ESP32. This connection enables the GPS module to send real-time location data such as latitude and longitude to the microcontroller.

Functional Description

The NEO-6M GPS module is responsible for providing the real-time geographical location of the vehicle. It continuously receives signals from GPS satellites and calculates parameters such as latitude and longitude. When an accident is detected by the ESP32, the GPS module supplies the exact coordinates of the incident location. This information is essential for emergency responders or contacts to quickly locate the accident site.

7. SIM800L GSM Module

Specifications:

- Operating Voltage: 3.7V – 4.2V (important)
- Supports GSM/GPRS communication
- Can send SMS, make calls, and use data services
- Controlled using AT commands
- Compact and low-cost module



Fig.3.6 SIM800L GSM Module

Connectivity

The SIM800L GSM module is also connected to the ESP32 using UART communication. Its TX pin is connected to the RX pin of the ESP32, and its RX pin is connected to the TX pin of the ESP32, sometimes through a voltage divider to ensure safe voltage levels. The module requires a dedicated external power supply of around 3.7V to 4.2V, as it cannot be powered directly from the ESP32. The ground of the SIM800L is connected to the common ground shared by all components. This configuration allows the ESP32 to control the GSM module and send SMS alerts during emergencies.

Functional Description

The SIM800L GSM module provides communication capabilities to the system. It is used to send SMS alerts to predefined phone numbers when an accident is detected. Upon receiving a command from the ESP32, the module sends a message containing the accident alert along with the GPS location link. This ensures that emergency contacts are immediately informed about the incident, enabling faster response and assistance.

System Implementation

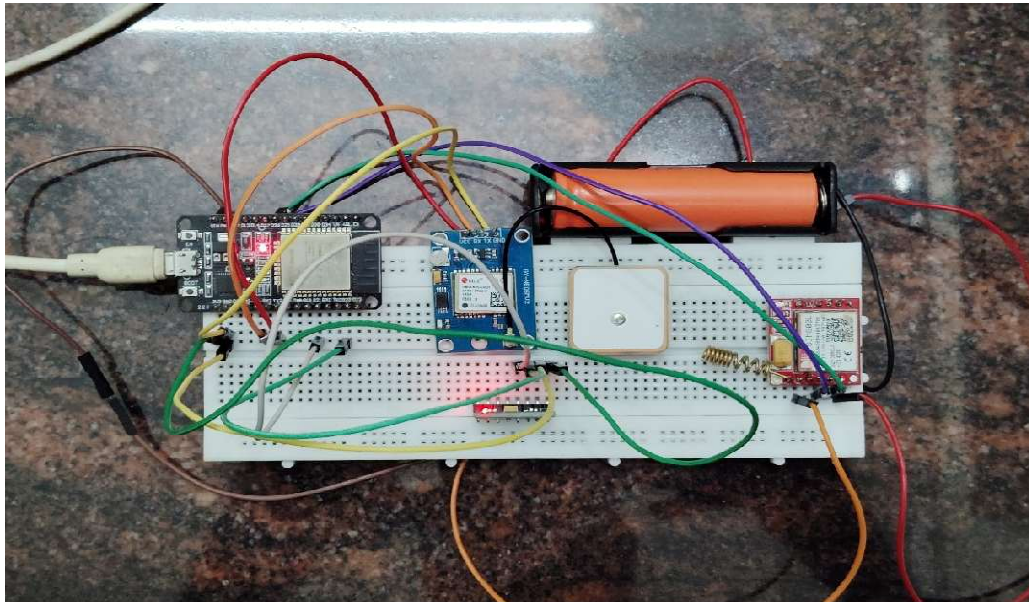


Fig 3.5 Hardware setup

Hardware Setup

The accident detection and alert system is implemented using an ESP32 microcontroller integrated with an MPU6050 sensor, a GPS module, and a GSM module. The ESP32 serves as the central processing unit, coordinating data acquisition, processing, and communication. The MPU6050 sensor is interfaced using I2C communication through the SDA and SCL pins, while the GPS and GSM modules are connected using UART communication. Special attention is

given to the power supply requirements, particularly for the GSM module, which requires a stable external voltage source to function reliably. The overall hardware setup ensures continuous monitoring of the vehicle's motion and enables real-time communication during emergency situations.

Accident Detection Algorithm

Initially, the system identifies a potential accident when both the acceleration and gyroscope magnitudes exceed predefined threshold values. Once such an event is detected, the system introduces a delay of a few seconds instead of immediately triggering an alert. This delay helps in filtering out temporary disturbances such as bumps, sudden braking, or minor shocks.

```
Accel (g): 1.05 | Gyro (deg/s): 34.68
Accel (g): 1.12 | Gyro (deg/s): 67.49
Accel (g): 2.30 | Gyro (deg/s): 269.57
Impact detected...
Accel (g): 2.90 | Gyro (deg/s): 433.24
Impact detected...
Accel (g): 0.92 | Gyro (deg/s): 51.59
Accel (g): 1.08 | Gyro (deg/s): 58.67
Accel (g): 1.11 | Gyro (deg/s): 75.10
Accel (g): 0.84 | Gyro (deg/s): 10.19
Accel (g): 1.15 | Gyro (deg/s): 22.99
Accel (g): 1.12 | Gyro (deg/s): 8.55
🚨 ACCIDENT DETECTED 🚨
```

Fig3.6 Serial Monitor Output

After the delay, the system checks whether the acceleration has returned to a stable range close to 1g. This stability indicates that the vehicle has come to rest after the impact. If this condition is satisfied, the system confirms that an actual accident has occurred. The observed output shows intermediate messages such as “Impact detected” followed by stable acceleration values and finally the

confirmation message “ACCIDENT DETECTED,” demonstrating the effectiveness of this multi-stage detection process.

SMS Alert Transmission

After acquiring the location, the GSM module is used to send an emergency alert message to a predefined mobile number. The communication with the GSM module is carried out using AT commands. The message includes a notification indicating that an accident has occurred along with the Google Maps link of the location. As observed in the output, the received SMS on the mobile phone contains the accident alert and a clickable link that opens the exact location on Google Maps, enabling immediate assistance and response.

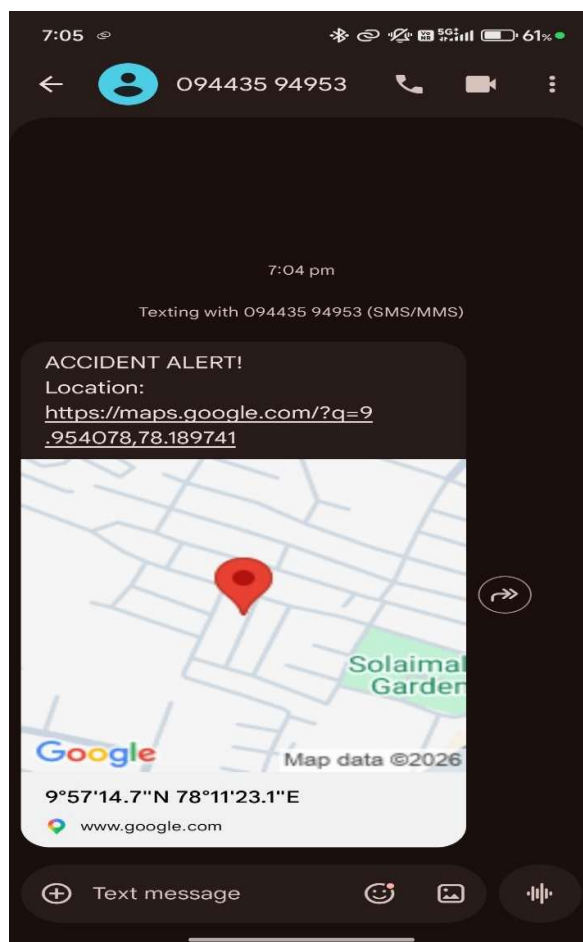


Fig.3.7 Alert Message

CHAPTER 4

RESULTS AND DISCUSSION

The IoT-based battery monitoring system using the ESP8266 module was successfully designed and implemented to monitor key parameters such as battery voltage, current, and overall health status in real time. The system collected data from sensors and transmitted it to a cloud-based IoT platform via Wi-Fi, allowing users to remotely view battery performance through a mobile or web dashboard. The results showed accurate and stable readings, with minimal error compared to conventional measuring instruments, proving the ESP8266's reliability as a low-cost controller for IoT applications. The system also provided visual indicators and alert notifications when the battery voltage dropped below a set threshold, helping to prevent over-discharge and extend battery life.

During operation, the Wi-Fi connectivity of the ESP8266 remained stable within a moderate range, and the IoT platform updated data at regular intervals to ensure real-time visibility of battery conditions. The discussion indicates that this system provides an efficient and scalable solution for smart energy management and remote battery supervision. Minor limitations were observed, such as small fluctuations in sensor readings due to electrical noise and dependence on a stable internet connection, which can be improved through calibration and data buffering.

Overall, the project demonstrates that integrating IoT technology with traditional battery systems enhances monitoring capabilities, promotes energy efficiency, and supports preventive maintenance. It also opens the door for future developments like adding temperature and charge-cycle monitoring, using predictive analysis for battery health, and integrating long-range communication modules such as LoRa or GSM for remote applications.

FUTURE SCOPE

Integration with Mobile Applications

The system can be enhanced by developing a dedicated mobile application that provides real-time accident alerts, live tracking, and user status updates. This would allow emergency contacts to monitor the rider's location continuously and respond.

Cloud and IoT Integration

Future improvements can include integration with cloud platforms to store accident data and provide real-time monitoring through IoT dashboards. This would enable data analysis, remote access, and improved coordination with emergency services.

Machine Learning for Improved Accuracy

Machine learning algorithms can be incorporated to analyze sensor data more intelligently and distinguish between actual accidents and false triggers such as sudden braking or road bumps. This will significantly improve the accuracy and reliability of the system.

Automatic Emergency Service Notification

Creating a custom mobile or web application allows real-time monitoring, data visualization, and alert notifications. Features such as analytics, historical performance graphs, and cloud dashboards improve accessibility and efficiency.

Addition of More Sensors

The system can be enhanced by integrating additional sensors such as pressure sensors, heart rate monitors, or alcohol detection sensors. This would provide more comprehensive safety monitoring for the rider.

CONCLUSION

The accident detection and alert system for two-wheelers using the ESP32 microcontroller provides an efficient and reliable solution to improve road safety. By integrating the MPU6050 sensor, the system is capable of continuously monitoring the motion and orientation of the vehicle and accurately detecting accident conditions such as sudden impacts or abnormal tilts. Upon detection, the system successfully retrieves the location using the NEO-6M GPS module and sends an immediate alert through the SIM800L GSM module to predefined emergency contacts.

The proposed system is compact, cost-effective, and suitable for real-time implementation in two-wheelers. It significantly reduces the delay in communication after an accident, thereby increasing the chances of timely medical assistance and survival. The inclusion of features such as automatic detection and alert transmission makes the system highly useful in remote or unattended areas.

Overall, this project demonstrates how embedded systems and communication technologies can be effectively utilized to address real-world safety challenges. With further improvements and enhancements, the system has strong potential for practical deployment and can contribute to reducing accident-related fatalities.

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